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Enantioselective addition of diethylzinc to aldehydes catalyzed by *o*-xylylene-type chiral 1,4-amino alcohols with an aминаl structure

Masatoshi Asami*, Ayano Hasome, Naoyuki Yachi, Naoya Hosoda, Yoshitaka Yamaguchi, Suguru Ito

Department of Advanced Materials Chemistry, Graduate School of Engineering, Yokohama National University, 79-5 Tokiwadai, Hodogaya-ku, Yokohama 240-8501, Japan

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ABSTRACT

A series of *o*-xylylene-type chiral 1,4-amino alcohols with an aминаl structure was synthesized starting from (S)-2-(arylaminoethyl)pyrrolidine, *o*-bromobenzaldehyde, and a diaryl ketone. The enantioselective addition of diethylzinc to aldehydes was examined by using the 1,4-amino alcohols, and the corresponding chiral secondary alcohols were obtained with high enantioselectivities (up to 98% ee).

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1. Introduction

The development of new chiral ligands and chiral catalysts has continuously been a main research subject in the field of asymmetric synthesis. Amino alcohols are well-investigated ligands for various asymmetric transformations.¹ Since the initial report by Oguni and Omi in 1984,² a number of chiral 1,2- and 1,3-amino alcohols have been investigated as chiral ligands to exhibit high levels of enantioinductions in the reaction of diethylzinc to aldehydes.^{3–5} Although the enantioselective addition of diethylzinc to aldehydes has been investigated as the benchmark reaction of chiral amino alcohol ligands, only a limited number of 1,4-amino alcohols are known as efficient chiral ligands in the reaction probably because of the difficulty in the enantioinduction due to the formation of relatively flexible seven-membered chelate structures generated from diethylzinc and 1,4-amino alcohol ligands.⁶

We have previously reported that chiral 1,4-amino alcohols with an *o*-xylylene structure were efficient chiral ligands in the

enantioselective addition of diethylzinc to aldehydes.⁷ Of the two benzylic carbons bearing amino or hydroxy groups in *o*-xylylene-type ligands, the absolute configuration of the stereogenic carbon with the amino group was found to play a critical role in determining the stereochemical outcome of the reaction. Our previous *o*-xylylene-type 1,4-amino alcohols were derived from (S,S)-1,2-bis(1-hydroxypropyl)benzene, prepared by the stepwise enantioselective addition of diethylzinc to aromatic aldehydes,^{7a,8} or commercially available (R)-1-phenylethylamine.^{7b} On the other hand, we previously reported the enantioselective addition using chiral 1,2-amino alcohol with an aминаl structure,⁹ in which the stereogenic center bearing the amino group was constructed by the diastereoselective formation of a chiral aминаl¹⁰ from phenylglyoxal and (S)-2-(anilinomethyl)pyrrolidine. These results prompted us to develop a new series of *o*-xylylene-type chiral 1,4-amino alcohol ligands **1** with a rigid structure from three components: (S)-proline-derived diamine **2**, *o*-bromobenzaldehyde **3**, and ketone **4** (Fig. 1); i.e., the diastereoselective aминаl formation

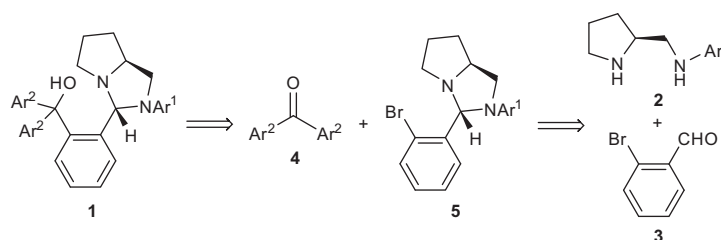


Figure 1. Design of *o*-xylylene-type chiral 1,4-amino alcohol with aминаl structure **1**.

* Corresponding author. Tel./fax: +81 45 339 3968.

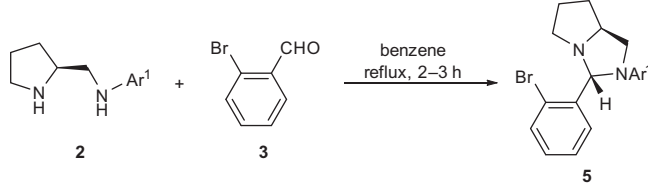
E-mail address: m-asami@ynu.ac.jp (M. Asami).

from diamine **2** and aldehyde **3** followed by lithiation of **5** and the addition of the lithiated species to ketone **4** would provide 1,4-amino alcohol **1**. This synthetic route would enable the facile synthesis of various 1,4-amino alcohols **1** by choosing the substituents of diamine **2** and ketone **4**. Herein we report the synthesis of a series of chiral 1,4-amino alcohols **1** and the enantioselective addition of diethylzinc to aldehydes using them.

2. Results and discussion

Three chiral amins **5a–c** were prepared from (*S*)-2-(arylaminomethyl)pyrrolidines **2a–c** and *o*-bromobenzaldehyde (**3**). (*S*)-2-(Anilinoethyl)pyrrolidine **2a**^{11a} and aldehyde **3** were refluxed in benzene with removal of water azeotropically for 2 h to give chiral aminal **5a**^{11b} in 86% yield after recrystallization (entry 1). Similarly, chiral aminal **5b** was synthesized from **3** and (*S*)-2-(*p*-anisidinomethyl)pyrrolidine **2b**^{11a} in 58% yield after recrystallization (entry 2). In the case of the reaction with (*S*)-2-(*p*-trifluoromethylanilino)methylpyrrolidine **2c**,^{11c} aminal **5c** was

Table 1
Diastereoselective synthesis of amins **5a–c**

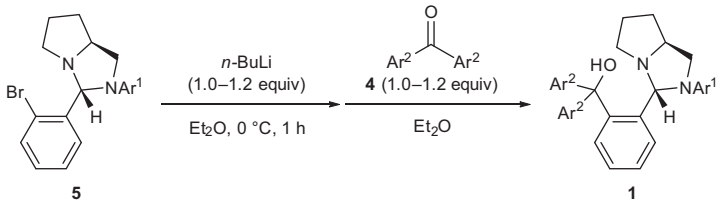


Entry	Amin	Ar ¹	Yield (%)
1	5a	Ph	86 ^a
2	5b	4-MeOC ₆ H ₄	58 ^a
3	5c	4-CF ₃ C ₆ H ₄	99 ^b

^a Isolated yield after recrystallization.

^b Yield of almost pure crude product.

Table 2
Synthesis of 1,4-amino alcohols **1a–k**



Entry	Amino alcohol	Ar ¹	Ar ²	Yield (%)
1	1a	Ph	Ph	94 ^a
2	1b	4-MeOC ₆ H ₄	Ph	37 ^b
3	1c	4-CF ₃ C ₆ H ₄	Ph	79 ^b
4	1d	Ph	4-MeC ₆ H ₄	67 ^b
5	1e	Ph	4-MeOC ₆ H ₄	51 ^b
6	1f	Ph	4-MeOC ₆ H ₄	81 ^b
7	1g	Ph	4-CF ₃ C ₆ H ₄	52 ^b
8	1h	Ph	3-CF ₃ C ₆ H ₄	99 ^a
9	1i	Ph	2-CF ₃ C ₆ H ₄	78 ^b
10	1j	Ph	3,5-(CF ₃) ₂ C ₆ H ₃	70 ^b
11	1k	Ph	3,4,5-F ₃ C ₆ H ₂	68 ^a
			C ₆ F ₅	27 ^b
				12 ^b

^a Isolated yield after column chromatography.

^b Isolated yield after recrystallization.

obtained quantitatively and the almost pure **5c** was used directly in the next step without further purification (entry 3). It should be noted that the aminal formation was highly diastereoselective and **5** was obtained in all cases (dr >20:1 by ¹H NMR).

1,4-Amino alcohols **1a–k** were synthesized from amins **5a–c** and various ketones **4** in which symmetrical ketones were used to avoid the formation of diastereomers (**Table 2**). Aminal **5a** was treated with butyllithium in diethyl ether at 0 °C for 1 h, after which the reaction of the lithiated species with benzophenone (Ar² = Ph) gave 1,4-amino alcohol **1a** in 94% yield after silica-gel column chromatography (entry 1). The structure of **1a** was confirmed by X-ray crystallographic analysis of a single crystal obtained by recrystallization from ethyl acetate (**Fig. 2**). The lithiated species of **5b** and **5c** were also treated with benzophenone to give the corresponding 1,4-amino alcohols **1b** and **1c** in 79% and 67% yields after recrystallization, respectively (entries 2 and 3). Various aminal-type 1,4-amino alcohols **1d–k** were also obtained from aminal **5a** and various symmetrical ketones **4** bearing electron-donating (–Me or –OMe) or -withdrawing (–CF₃ or –F) groups on the benzene ring. The crude products of **1d–k** were recrystallized directly or after silica-gel column chromatography to give pure products in 12–81% yields (entries 4–11). The recrystallized 1,4-amino alcohols **1a–k** were used as chiral ligands in the following experiments.

The enantioselective addition of diethylzinc (2.0 equiv) to benzaldehyde was examined using 1,4-amino alcohol **1a** (**Table 3**). In the presence of 10 mol % of **1a**, the reaction was carried out in toluene at room temperature for 7 h to give (*S*)-1-phenyl-1-propanol in 92% yield and with 92% ee (entry 1). The effect of the solvent was investigated by using hexane, cyclohexane, diethyl ether (Et₂O), tetrahydrofuran (THF), or dichloromethane (CH₂Cl₂) instead of toluene (entries 2–6). Toluene was found to be the best solvent in terms of both yield and enantioselectivity of the product. The yield and enantioselectivity were not improved upon when the reaction was carried out at 40 °C or 0 °C (entries 7 and 8). Increasing the amount of catalyst (15 or 20 mol %) was also ineffective in terms of improving the enantioselectivity (entries 9 and 10). Meanwhile,

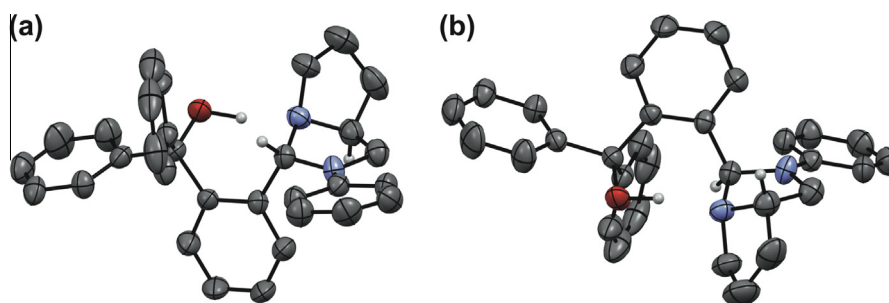


Figure 2. ORTEP structures of 1,4-amino alcohol **1a** with thermal ellipsoids shown at 50% probability (C = gray, O = red, N = blue, H = white). All hydrogen atoms except those of two stereogenic carbons and the hydroxy group are omitted for the sake of clarity. (a) Front view. (b) Back view.

Table 3

Enantioselective addition of diethylzinc to benzaldehyde catalyzed by 1,4-amino alcohol **1a**

Entry	Solvent	1a (mol %)	Yield (%)	Ee ^a (%)
1	Toluene	10	92	92
2	Hexane	10	86	90
3	Cyclohexane	10	86	82
4	Et ₂ O	10	86	91
5	THF	10	38	86
6	CH ₂ Cl ₂	10	62	88
7 ^b	Toluene	10	86	88
8 ^c	Toluene	10	47	91
9	Toluene	15	93	91
10	Toluene	20	91	89
11	Toluene	5	66	90

^a Determined by HPLC analysis.

^b Reaction was carried out at 40 °C.

^c Reaction was carried out at 0 °C.

the yield and enantioselectivity decreased slightly when using a smaller amount of **1a** (5 mol %, entry 11). As the best result was obtained in entry 1, the following experiments were carried out in toluene at room temperature with 10 mol % of **1**.

Table 4

Enantioselective addition of diethylzinc to benzaldehyde catalyzed by 1,4-amino alcohols **1a–c**

Entry	Amino alcohol	Ar ¹	Yield (%)	Ee ^a (%)
1	1a	Ph	90	92
2	1b	4-MeOC ₆ H ₄	62	92
3	1c	4-CF ₃ C ₆ H ₄	86	92

^a Determined by HPLC analysis.

The substituent effect of the benzene ring (Ar¹) attached to the nitrogen atom of the aminal moiety was next examined (Table 4). When 1,4-amino alcohol **1b** (Ar¹ = 4-MeOC₆H₄) bearing an electron-donating group on the benzene ring was used in the reaction of benzaldehyde with diethylzinc, (*S*)-1-phenyl-1-propanol was obtained in 62% yield and with 92% ee (entry 2). The product was obtained in 86% yield and with 92% ee by using **1c** (Ar¹ = 4-CF₃C₆H₄) with an electron-withdrawing group on the benzene ring (entry 3). The enantioselectivity was not affected by the substituent of Ar¹, and the highest yield was observed by using non-substituted **1a**.

The effect of substituents on the benzene ring (Ar²), attached to the benzylic carbon, was also studied in the enantioselective reaction. The results are summarized in Table 5. The yield and enantioselectivity slightly decreased when using **1d** (Ar² = 4-MeC₆H₄) and **1e** (Ar² = 4-MeOC₆H₄) with electron-donating groups on the *p*-position of the benzene ring (entries 1 and 2). In contrast, (*S*)-1-phenyl-1-propanol was obtained in 92% yield and with 97% ee by using **1f** (Ar² = 4-CF₃C₆H₄) bearing an electron-withdrawing group at the *p*-position (entry 3). When 1,4-amino alcohol **1g** (Ar² = 3-CF₃C₆H₄) was used as the chiral ligand, the reaction was completed within 5 h to give the corresponding chiral secondary alcohol in 94% yield and with 97% ee (entry 4). The yield was decreased to 69% when *o*-trifluoromethyl-substituted **1h** was used in the reaction, probably due to the steric hindrance of the substituent (entry 5). Although high enantioselectivities (97% ee) were

Table 5

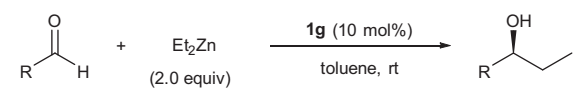
Enantioselective addition of diethylzinc to benzaldehyde catalyzed by 1,4-amino alcohols **1d–k**

Entry	Amino alcohol	Ar ²	Yield (%)	Ee ^a (%)
1	1d	4-MeC ₆ H ₄	82	91
2	1e	4-MeOC ₆ H ₄	77	91
3	1f	4-CF ₃ C ₆ H ₄	92	97
4 ^b	1g	3-CF ₃ C ₆ H ₄	94	97
5	1h	2-CF ₃ C ₆ H ₄	69	93
6	1i	3,5-(CF ₃) ₂ C ₆ H ₃	77	97
7	1j	3,4,5-F ₃ C ₆ H ₂	83	97
8	1k	C ₆ F ₅	54	94

^a Determined by HPLC analysis.

^b Reaction was carried out for 5 h.

Table 6
Enantioselective addition of diethylzinc to various aldehydes catalyzed by 1,4-amino alcohol **1g**



Entry	R	Time (h)	Yield (%)	Ee ^a (%)
1 ^b	Ph	5	94	97
2 ^c	2-BrC ₆ H ₄	18	76	97
3 ^c	1-Naphthyl	18	73	97
4	2-MeOC ₆ H ₄	5	91	96
5	4-MeOC ₆ H ₄	5	83	98
6	c-C ₆ H ₁₁	5	52	79
7	(<i>E</i>)-PhCH=CH	5	88	67
8 ^c	PhCH ₂ CH ₂	18	73	38
9 ^d	Ph	5	85	98

^a Determined by HPLC analysis.

^b Table 5, entry 4.

^c The reaction was not completed in 5 h.

^d Me₂Zn was used instead of Et₂Zn.

also achieved when using **1i** (Ar² = 3,5-(CF₃)₂C₆H₃) and **1j** (Ar² = 3,4,5-F₃C₆H₂), the yields of the product were lower than that of the reaction using **1g** (entries 6 and 7). The use of **1k** bearing perfluorophenyl groups gave the product in low yield (entry 8).

Since the best result was obtained by using 1,4-amino alcohol **1g**, the reaction with various aldehydes was examined (Table 6). High enantioselectivities (96–98% ee) were attained in the reaction with aromatic aldehydes: 2-bromobenzaldehyde, 1-naphthaldehyde, 2-anisaldehyde, and 4-anisaldehyde (entries 2–5). These selectivities were higher than those using our previous *o*-xylylene-type 1,4-amino alcohols.⁷ The reactions with cyclohexanecarboxaldehyde, (*E*)-cinnamaldehyde, and 3-phenylpropanal afforded the corresponding chiral secondary alcohols in moderate to good yields and enantioselectivities (entries 6–8). The reaction of benzaldehyde with dimethylzinc was also carried out, which resulted in the formation of (*S*)-1-phenyl-1-ethanol in 85% yield and with 98% ee (entry 9).

The mechanism of the enantioselective addition of organozinc compounds to aldehydes using amino alcohol ligands has been well investigated computationally.¹² Based on the literature review, we have proposed a stereochemical course of the reaction

using 1,4-amino alcohol **1** as shown in Figure 3. Initially, zinc complex **A** with a seven-membered chelate structure is generated from amino alcohol ligand **1** and diethylzinc. The in situ formed complex **A** acts as a catalyst in the enantioselective addition of diethylzinc to aldehydes. Next, another diethylzinc and aldehyde approach to catalyst **A** from the less hindered side to form a transition structure **B**. At this stage, a pseudo-boat conformation is considered to be the most suitable structure, in which the arylamino group of amina moiety occupies the pseudo-equatorial position. The stability of this conformation is supported by the crystalline-state structure of 1,4-amino alcohol **1a** (Fig. 2b). In addition, the R substituent of the aldehyde prevents steric repulsion with the catalyst. Therefore, the corresponding alcohol should be produced with an (*S*)-configuration from transition structure **B**.

3. Conclusion

In conclusion, we have designed and synthesized a series of chiral 1,4-amino alcohol ligands **1a–k**, which consist of rigid amina and *o*-xylylene structures. The 1,4-amino alcohol ligands were used in the enantioselective addition of diethylzinc to aldehydes, and the best result was obtained by using **1g** bearing 3-trifluoromethylphenyl groups. In the presence of **1g**, high enantioselectivities were achieved especially in reactions using aromatic aldehydes. By taking advantage of their rigid structures, the 1,4-amino alcohols developed herein could be applicable to various asymmetric transformations as efficient chiral ligands and chiral catalysts.

4. Experimental

4.1. General

All air-sensitive experiments were carried out under an atmosphere of argon unless otherwise noted. IR spectra were recorded on a HORIBA FT-730 spectrometer. ¹H and ¹³C NMR spectra were recorded on Bruker DRX-300, JEOL ECX-400, or Bruker DRX-500 spectrometer using tetramethylsilane as an internal standard. Optical rotations were measured on a JASCO P-1000 automatic polarimeter. HPLC analyses were carried out with JASCO instruments (pump, PU-2080 plus; detector, UV-2075). Enantiomeric excesses were determined by HPLC using Daicel Chiralcel OD-H, OB, or AS-H (25 cm × 0.46 cm i.d.) column. Elemental analyses were carried out on a Vario EL III Elemental analyzer. High-resolution mass spectra (HRMS) were recorded on a Hitachi Nano Frontier LD spectrometer. TLC analyses were done on silica-gel 60 F₂₅₄-precoated aluminum backed sheets (E. Merck). Preparative TLC separations were performed on silica-gel-coated plates (Wakogel B-5F, 20 cm × 20 cm). Wakogel C-200, Silica gel 60 N (spherical, neutral, 63–210 μm), and Chromatorex NH (DM1020, Fuji Silysia Chemical Ltd, Japan) were used for column chromatography. A hexane solution of diethylzinc (1.1 M, Kanto Chemical Co., Inc., Japan) and a hexane solution of dimethylzinc (1.0 M, Kanto Chemical Co., Inc., Japan) were used for the enantioselective addition. Diethyl ether (dehydrated) and tetrahydrofuran (dehydrated, stabilizer free) were purchased from Kanto Chemical Co., Inc. Other solvents were purified and dried according to standard procedures.

4.2. Synthesis of amins 5a–c

4.2.1. (3*R*,7*aS*)-3-(2-Bromophenyl)-2-phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazole **5a**^{11b}

A benzene (110 mL) solution of (*S*)-2-(anilinomethyl)pyrrolidine (10.34 g, 55.0 mmol) and *o*-bromobenzaldehyde (9.69 g,

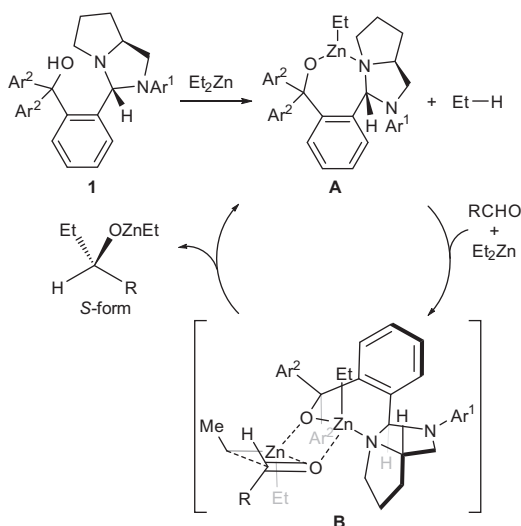


Figure 3. Proposed stereochemical course of the reaction using 1,4-amino alcohol **1**.

55.0 mmol) was heated at reflux with removal of water azeotropically for 2 h. After removal of the solvent under reduced pressure, the crude product was recrystallized from cyclohexane to give amina **5a** (16.2 g, 86%) as colorless crystals. Mp: 169–170 °C (lit. 169–170.5 °C); $[\alpha]_D^{23} = +124$ (c 1.0, CH₂Cl₂), lit. $[\alpha]_D^{23} = +125$ (c 1.01, CH₂Cl₂); IR (KBr): $\nu_{\max}$ 3050, 2967, 2913, 2874, 2834, 1597, 1568, 1505, 1486, 1472, 1442, 1372, 1265, 1202, 1159, 1112, 1095, 1026, 994, 935, 905, 883, 836, 765, 748 cm⁻¹; ¹H NMR (400 MHz, CDCl₃): δ (ppm) 7.62 (d, *J* = 7.7 Hz, 1H), 7.10–7.20 (m, 5H), 6.68 (t, *J* = 7.3 Hz, 1H), 6.38 (d, *J* = 8.2 Hz, 2H), 5.64 (s, 1H), 3.91 (q, *J* = 8.3 Hz, 1H), 3.79 (t, *J* = 8.3 Hz, 1H), 3.52–3.58 (m, 1H), 3.28 (t, *J* = 8.3 Hz, 1H), 2.84 (q, *J* = 8.7 Hz, 1H), 2.10–2.18 (m, 1H), 1.87–2.05 (m, 3H); ¹³C NMR (126 MHz, CDCl₃): δ (ppm) 145.7, 140.2, 133.5, 129.14, 129.12, 127.5, 127.3, 123.7, 116.5, 112.3, 83.0, 60.6, 53.9, 53.2, 27.9, 24.0.

4.2.2. (3*R*,7*S*)-3-(2-Bromophenyl)-2-(4-methoxyphenyl)hexahydro-1*H*-pyrrolo[1,2-*c*]imidazole **5b**

Brown solid; mp: 118–119 °C; $[\alpha]_D^{19} = +96.2$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3037, 3010, 2949, 2908, 2831, 2810, 1619, 1519, 1510, 1487, 1461, 1441, 1364, 1351, 1330, 1259, 1237, 1196, 1182, 1160, 1112, 1094, 1040, 1017, 983, 934, 910, 884, 810, 761 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 7.61 (dd, *J* = 7.7, 0.9 Hz, 1H), 7.08–7.26 (m, 3H), 6.74–6.80 (m, 2H), 6.29–6.35 (m, 2H), 5.56 (s, 1H), 3.86–3.95 (m, 1H), 3.73–3.80 (m, 1H), 3.71 (s, 3H), 3.47–3.58 (m, 1H), 3.23 (t, *J* = 8.7 Hz, 1H), 2.84 (q, *J* = 8.9 Hz, 1H), 1.81–2.21 (m, 4H); ¹³C NMR (126 MHz, CDCl₃): δ (ppm) 151.3, 140.60, 140.56, 133.4, 129.1, 127.6, 127.5, 123.7, 114.9, 113.0, 83.5, 60.9, 55.8, 54.0, 53.7, 27.9, 24.0; HRMS-ESI (*m/z*): [M+H]⁺ Calcd for C₁₉H₂₂BrN₂O, 373.0910; Found, 373.0913.

4.2.3. (3*R*,7*S*)-3-(2-Bromophenyl)-2-(4-(trifluoromethyl)phenyl)-hexahydro-1*H*-pyrrolo[1,2-*c*]imidazole **5c**

White solid; mp: 43–44 °C; $[\alpha]_D^{19} = +86.7$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3057, 2968, 2927, 2877, 2844, 1615, 1571, 1531, 1462, 1440, 1379, 1328, 1262, 1193, 1155, 1111, 1069, 1022, 980, 934, 818, 754 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 7.63 (dd, *J* = 7.5, 0.9 Hz, 1H), 7.37 (d, *J* = 8.7 Hz, 2H), 7.04–7.23 (m, 3H), 6.37 (d, *J* = 8.7 Hz, 2H), 5.69 (s, 1H), 3.91 (q, *J* = 8.0 Hz, 1H), 3.80 (t, *J* = 8.0 Hz, 1H), 3.52–3.59 (m, 1H), 3.29 (t, *J* = 8.0 Hz, 1H), 2.81 (q, *J* = 8.8 Hz, 1H), 1.84–2.20 (m, 4H); ¹³C NMR (100 MHz, CDCl₃): δ (ppm) 147.5, 139.0, 133.4, 129.2, 127.4, 126.6, 126.2 (q, ³*J*_{C-F} = 3.4 Hz), 124.8 (q, ¹*J*_{C-F} = 270 Hz), 123.5, 117.9 (q, ²*J*_{C-F} = 33 Hz), 111.5, 82.7, 60.3, 53.6, 52.8, 27.7, 23.7; HRMS-ESI (*m/z*): [M+H]⁺ Calcd for C₁₉H₁₉BrF₃N₂, 411.0678; Found, 411.0687.

4.3. Synthesis of 1,4-amino alcohols **1a–k**

4.3.1. Diphenyl(2-((3*R*,7*S*)-2-phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)methanol **1a**

To a stirred solution of amina **5a** (1.03 g, 3.0 mmol) in Et₂O (5.0 mL), a hexane solution of *n*-BuLi (2.6 M, 1.4 mL) was added dropwise through a syringe at 0 °C. The reaction mixture was stirred at 0 °C for 1 h, after which the mixture was cooled to –78 °C. To the mixture was added dropwise an Et₂O (3.0 mL) solution of benzophenone (657 mg, 3.6 mmol) at –78 °C and the mixture was gradually warmed to room temperature and stirred overnight. Saturated aqueous ammonium chloride solution and water were then added to the reaction mixture. The aqueous layer was separated and the organic layer was extracted with CH₂Cl₂ three times. The combined organic layer was washed with water and brine, and dried over anhydrous magnesium sulfate. After the removal of solvent under reduced pressure, the crude product was purified by silica-gel column chromatography (Chromatorex NH, hexane/ethyl acetate = 8:1) and successively washed with Et₂O to give

1,4-amino alcohol **1a** (1.26 g, 94%) as a white solid. Recrystallization from ethyl acetate (31 mL) afforded colorless crystals of **1a** (490 mg, 37%). Mp: 178–180 °C; $[\alpha]_D^{28} = +292.9$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3447, 3060, 2971, 2938, 2879, 2832, 1595, 1504, 1368, 1355, 1288, 1188, 1083, 1035, 997, 916, 839, 750, 690 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 9.69 (br s, 1H), 7.55–7.64 (m, 2H), 7.25–7.50 (m, 8H), 7.08–7.16 (m, 5H), 6.74–6.79 (m, 1H), 6.65 (t, *J* = 7.2 Hz, 1H), 6.14 (d, *J* = 7.9 Hz, 2H), 5.01 (s, 1H), 3.77 (q, *J* = 7.8 Hz, 1H), 3.58 (t, *J* = 7.8 Hz, 1H), 3.15–3.26 (m, 2H), 2.19 (q, *J* = 9.2 Hz, 1H), 1.82–2.11 (m, 4H); ¹³C NMR (100 MHz, CDCl₃): δ (ppm) 148.1, 146.9, 146.6, 145.6, 137.4, 130.6, 129.1, 128.3, 128.1, 128.0, 127.7, 127.6, 127.5, 127.3, 127.2, 126.7, 116.2, 111.6, 81.6, 79.9, 60.5, 51.5, 50.6, 27.6, 22.9; Anal. Calcd for C₃₁H₃₀N₂O: C, 83.37; H, 6.77; N, 6.27. Found: C, 83.23; H, 6.71; N, 6.24.

4.3.2. (2-((3*R*,7*S*)-2-(4-Methoxyphenyl)hexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)diphenylmethanol **1b**

White solid; mp: 137–141 °C; $[\alpha]_D^{23} = +210.4$ (c 1.1, CHCl₃); IR (KBr): $\nu_{\max}$ 3445, 3055, 2936, 2831, 1511, 1489, 1450, 1348, 1238, 1176, 1036, 819, 763, 756, 710, 701 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 9.74 (br s, 1H), 7.59 (d, *J* = 7.3 Hz, 2H), 7.45 (d, *J* = 7.3 Hz, 2H), 7.25–7.42 (m, 6H), 7.06–7.18 (m, 3H), 6.69–6.80 (m, 3H), 6.04–6.10 (m, 2H), 4.95 (s, 1H), 3.70–3.81 (m, 1H), 3.72 (s, 3H), 3.56 (t, *J* = 8.0 Hz, 1H), 3.11–3.26 (m, 2H), 2.13–2.24 (m, 1H), 2.01–2.12 (m, 1H), 1.81–1.99 (m, 3H); ¹³C NMR (75.5 MHz, CDCl₃): δ (ppm) 151.0, 148.1, 147.0, 146.7, 140.4, 137.8, 130.5, 128.3, 128.2, 128.0, 127.7, 127.6, 127.4, 127.3 (×2), 126.7, 114.9, 112.0, 81.6, 80.2, 60.7, 55.8, 51.9, 50.6, 27.6, 22.9; Anal. Calcd for C₃₂H₃₂N₂O₂: C, 80.64; H, 6.77; N, 5.88. Found: C, 80.33; H, 6.76; N, 5.83.

4.3.3. Diphenyl(2-((3*R*,7*S*)-2-(4-(trifluoromethyl)phenyl)hexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)methanol **1c**

Pale yellow solid; mp: 161–162 °C; $[\alpha]_D^{23} = +227.0$ (c 1.3, CHCl₃); IR (KBr): $\nu_{\max}$ 3442, 3062, 2955, 2850, 1616, 1574, 1533, 1490, 1448, 1383, 1330, 1196, 1156, 1112, 1070, 1027, 983, 939, 911, 819, 763, 702 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 9.38 (br s, 1H), 7.60 (d, *J* = 7.5 Hz, 2H), 7.21–7.49 (m, 10H), 7.07–7.19 (m, 2H), 6.95–7.07 (m, 1H), 6.73–6.85 (m, 1H), 5.95–6.32 (m, 2H), 5.05 (s, 1H), 3.69–3.88 (m, 1H), 3.61 (t, *J* = 8.8 Hz, 1H), 3.23–3.36 (m, 1H), 3.18 (t, *J* = 8.8 Hz, 1H), 1.78–2.27 (m, 5H); ¹³C NMR (100 MHz, CDCl₃): δ (ppm) 147.9, 147.8, 147.0, 146.4, 136.5, 130.8, 128.3, 128.2, 127.9, 127.8, 127.6, 127.53, 127.48, 127.2, 126.8, 126.4 (q, ³*J*_{C-F} = 3.8 Hz), 125.0 (q, ¹*J*_{C-F} = 270 Hz), 118.0 (q, ²*J*_{C-F} = 33 Hz), 111.2, 81.6, 80.1, 60.4, 51.7, 50.7, 27.6, 22.9; Anal. Calcd for C₃₂H₂₉F₃N₂O: C, 74.69; H, 5.68; N, 5.44. Found: C, 74.49; H, 5.69; N, 5.40.

4.3.4. (2-((3*R*,7*S*)-2-Phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)di-*p*-tolylmethanol **1d**

White solid; mp: 194–198 °C; $[\alpha]_D^{28} = +281.0$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3449, 3060, 3026, 2952, 2924, 2856, 1600, 1573, 1505, 1473, 1450, 1370, 1312, 1176, 1161, 1085, 1043, 1021, 996, 937, 918, 837, 816, 796, 769, 746 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 9.56 (br s, 1H), 7.46 (d, *J* = 7.9 Hz, 2H), 7.33 (d, *J* = 8.3 Hz, 2H), 7.07–7.22 (m, 9H), 6.76–6.84 (m, 1H), 6.65 (t, *J* = 7.1 Hz, 1H), 6.15 (d, *J* = 7.9 Hz, 2H), 5.08 (s, 1H), 3.76 (q, *J* = 8.0 Hz, 1H), 3.58 (t, *J* = 8.0 Hz, 1H), 3.11–3.29 (m, 2H), 2.34 (s, 3H), 2.32 (s, 3H), 2.21 (q, *J* = 9.2 Hz, 1H), 1.80–2.12 (m, 4H); ¹³C NMR (100 MHz, CDCl₃): δ (ppm) 147.2, 145.6, 145.2, 144.0, 137.4, 136.7, 136.1, 130.5, 129.0, 128.8, 128.4, 128.1, 127.9, 127.6, 127.4, 127.1, 116.1, 111.7, 81.3, 79.9, 60.4, 51.6, 50.6, 27.6, 22.9, 21.1, 21.0; Anal. Calcd for C₃₃H₃₄F₆N₂O: C, 83.51; H, 7.22; N, 5.90. Found: C, 83.38; H, 7.32; N, 5.87.

4.3.5. Bis(4-methoxyphenyl)(2-((3*R*,7*aS*)-2-phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)methanol 1e

White solid; mp: 140–145 °C; $[\alpha]_D^{23} = +281.7$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3444, 3060, 2952, 2835, 1605, 1507, 1464, 1369, 1299, 1247, 1171, 1035, 839, 750 cm⁻¹; ¹H NMR (400 MHz, CDCl₃): δ (ppm) 9.55 (br s, 1H), 7.47 (d, *J* = 8.7 Hz, 2H), 7.33 (d, *J* = 8.2 Hz, 2H), 7.05–7.15 (m, 5H), 6.88–6.94 (m, 2H), 6.83–6.88 (m, 2H), 6.74–6.80 (m, 1H), 6.65 (t, *J* = 7.3 Hz, 1H), 6.15 (d, *J* = 8.2 Hz, 2H), 5.08 (s, 1H), 3.81 (s, 3H), 3.78 (s, 3H), 3.71–3.80 (m, 1H), 3.58 (t, *J* = 8.2 Hz, 1H), 3.15–3.26 (m, 2H), 2.23 (q, *J* = 9.2 Hz, 1H), 2.01–2.12 (m, 1H), 1.82–1.99 (m, 3H); ¹³C NMR (100 MHz, CDCl₃): δ (ppm) 158.7, 158.2, 147.4, 145.6, 140.3, 139.4, 137.4, 130.5, 129.4, 129.1, 128.3, 128.0, 127.6, 127.4, 116.2, 113.4, 113.0, 111.7, 81.1, 80.0, 79.9, 60.5, 55.2, 51.6, 50.7, 27.6, 22.9; Anal. Calcd for C₃₃H₃₄N₂O₃: C, 78.23; H, 6.76; N, 5.53. Found: C, 77.99; H, 6.86; N, 5.42.

4.3.6. (2-((3*R*,7*aS*)-2-Phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)bis(4-(trifluoromethyl)phenyl)methanol 1f

Pale yellow solid; mp: 179–183 °C; $[\alpha]_D^{28} = +241.6$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3448, 3066, 2975, 2880, 2840, 2642, 1613, 1602, 1504, 1471, 1409, 1367, 1327, 1167, 1122, 1067, 1017, 997, 847, 764, 746 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 10.05 (br s, 1H), 7.65–7.77 (m, 4H), 7.53–7.65 (m, 4H), 7.08–7.24 (m, 5H), 6.64–6.73 (m, 2H), 6.08 (d, *J* = 7.9 Hz, 2H), 4.93 (s, 1H), 3.71–3.82 (m, 1H), 3.61 (t, *J* = 8.3 Hz, 1H), 3.12–3.25 (m, 2H), 2.18–2.30 (m, 1H), 1.85–2.14 (m, 4H); ¹³C NMR (126 MHz, CDCl₃): δ (ppm) 152.0, 149.9, 145.4, 145.2, 137.4, 130.4, 130.0 (q, ²*J*_{C-F} = 32 Hz), 129.26 (q, ²*J*_{C-F} = 32 Hz), 129.23, 128.7, 128.5, 128.2, 128.0, 127.6, 125.3 (q, ³*J*_{C-F} = 3.7 Hz), 124.9 (q, ³*J*_{C-F} = 3.7 Hz), 124.2 (q, ¹*J*_{C-F} = 272 Hz), 124.0 (q, ¹*J*_{C-F} = 272 Hz), 116.7, 111.5, 81.3, 79.9, 60.6, 51.6, 50.7, 27.6, 22.9; Anal. Calcd for C₃₃H₂₈F₆N₂O: C, 68.03; H, 4.84; N, 4.81. Found: C, 67.94; H, 4.91; N, 4.71.

4.3.7. (2-((3*R*,7*aS*)-2-Phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)bis(3-(trifluoromethyl)phenyl)methanol 1g

White solid; mp: 173–174 °C; $[\alpha]_D^{28} = +253.3$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3446, 3062, 2950, 2843, 1597, 1505, 1491, 1430, 1368, 1328, 1286, 1164, 1134, 1074, 796, 744, 718, 703 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 10.21 (br s, 1H), 7.96 (br s, 2H), 7.73 (br s, 1H), 7.52–7.64 (m, 3H), 7.39–7.48 (br m, 2H), 7.09–7.22 (m, 5H), 6.64–6.72 (m, 2H), 6.10 (d, *J* = 7.9 Hz, 2H), 4.83 (s, 1H), 3.71–3.82 (m, 1H), 3.59 (t, *J* = 8.3 Hz, 1H), 3.10–3.26 (m, 2H), 2.16–2.27 (m, 1H), 1.85–2.14 (m, 4H); ¹³C NMR (126 MHz, CDCl₃, 50 °C): δ (ppm) 149.4, 147.4, 145.6, 145.5, 137.4, 131.8, 131.1 (q, ²*J*_{C-F} = 32 Hz), 130.68, 130.67 (q, ²*J*_{C-F} = 32 Hz), 130.5, 129.3, 128.9, 128.7, 128.23, 128.21, 128.1, 125.1 (q, ³*J*_{C-F} = 3.7 Hz), 124.6 (q, ³*J*_{C-F} = 3.7 Hz), 124.4 (q, ¹*J*_{C-F} = 272 Hz), 124.2 (q, ¹*J*_{C-F} = 272 Hz), 124.0 (×2, q, ³*J*_{C-F} = 3.7 Hz), 116.8, 111.6, 81.3, 80.2, 60.7, 51.6, 50.7, 27.8, 22.9; Anal. Calcd for C₃₃H₂₈F₆N₂O: C, 68.03; H, 4.84; N, 4.81. Found: C, 68.15; H, 4.89; N, 4.74.

4.3.8. (2-((3*R*,7*aS*)-2-Phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)bis(2-(trifluoromethyl)phenyl)methanol 1h

Pale yellow solid; mp: 204–210 °C; $[\alpha]_D^{28} = +199.8$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3441, 3061, 2940, 2804, 1600, 1505, 1492, 1444, 1366, 1306, 1274, 1162, 1137, 1107, 1060, 1035, 996, 938, 918, 765, 744 cm⁻¹; ¹H NMR (500 MHz, CDCl₃): δ (ppm) 9.31 (br s, 1H), 7.84–7.88 (m, 2H), 7.43–7.48 (m, 2H), 7.35 (t, *J* = 7.7 Hz, 1H), 7.26 (t, *J* = 7.7 Hz, 1H), 7.05–7.17 (m, 5H), 6.94–6.98 (m, 1H), 6.73 (d, *J* = 7.9 Hz, 1H), 6.60–6.67 (m, 2H), 6.04 (d, *J* = 7.9 Hz, 2H), 4.95 (s, 1H), 3.74 (q, *J* = 8.3 Hz, 1H), 3.59 (t, *J* = 8.3 Hz, 1H), 3.27–3.33 (m, 1H), 3.24 (t, *J* = 8.3 Hz, 1H), 2.29 (q, *J* = 9.1 Hz, 1H), 1.99–2.07 (m, 1H), 1.79–1.93 (m, 3H); ¹³C NMR (126 MHz, CDCl₃): δ (ppm) 147.8, 147.4, 146.4, 145.5, 137.3, 130.7, 130.52 (q, ²*J*_{C-F} = 32 Hz), 130.48, 130.3 (×2), 130.1, 129.49 (q, ³*J*_{C-F} = 6.4 Hz),

129.48 (q, ²*J*_{C-F} = 32 Hz), 129.0, 128.8 (q, ³*J*_{C-F} = 6.4 Hz), 128.1, 128.0, 127.8, 127.5, 127.1, 124.7 (q, ¹*J*_{C-F} = 274 Hz), 124.3 (q, ¹*J*_{C-F} = 274 Hz), 116.2, 111.6, 84.6, 80.3, 60.1, 52.2, 50.7, 27.7, 22.9; Anal. Calcd for C₃₃H₂₈F₆N₂O: C, 68.03; H, 4.84; N, 4.81. Found: C, 68.10; H, 4.95; N, 4.74.

4.3.9. Bis(3,5-bis(trifluoromethyl)phenyl)(2-((3*R*,7*aS*)-2-phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)methanol 1i

White solid; mp: 130–133 °C; $[\alpha]_D^{28} = +205.0$ (c 0.47, CHCl₃); IR (KBr): $\nu_{\max}$ 3424, 3065, 2935, 2853, 2784, 2644, 1602, 1506, 1464, 1367, 1281, 1174, 1138, 998, 898, 844, 747, 715, 683 cm⁻¹; ¹H NMR (300 MHz, CDCl₃): δ (ppm) 10.69 (s, 1H), 8.11 (s, 2H), 7.83–7.97 (m, 4H), 7.11–7.31 (m, 5H), 6.71 (t, *J* = 7.4 Hz, 1H), 6.57–6.62 (m, 1H), 6.06 (d, *J* = 8.3 Hz, 2H), 4.67 (s, 1H), 3.70–3.81 (m, 1H), 3.60 (t, *J* = 8.6 Hz, 1H), 3.24 (t, *J* = 8.6 Hz, 1H), 3.03–3.13 (m, 1H), 2.17–2.30 (m, 1H), 1.88–2.17 (m, 4H); ¹³C NMR (126 MHz, CDCl₃): δ (ppm) 150.7, 148.0, 145.1, 143.8, 136.9, 132.2 (q, ²*J*_{C-F} = 33 Hz), 131.6 (q, ²*J*_{C-F} = 33 Hz), 130.2, 129.5, 129.3, 129.0, 128.6, 128.2–128.4 (m), 127.0–127.2 (m), 123.3 (q, ¹*J*_{C-F} = 273 Hz), 123.2 (q, ¹*J*_{C-F} = 273 Hz), 122.0–122.2 (m), 121.5–121.7 (m), 117.1, 111.5, 80.9, 80.0, 60.8, 51.2, 50.5, 27.6, 22.8; Anal. Calcd for C₃₅H₂₆F₁₂N₂O: C, 58.50; H, 3.65; N, 3.90. Found: C, 58.47; H, 3.84; N, 3.80.

4.3.10. (2-((3*R*,7*aS*)-2-Phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)bis(3,4,5-trifluorophenyl)methanol 1j

Pale yellow solid; mp: 184–185 °C; $[\alpha]_D^{21} = +215.0$ (c 1.0, CHCl₃); IR (KBr): $\nu_{\max}$ 3435, 3067, 2947, 2846, 1620, 1597, 1522, 1506, 1433, 1367, 1340, 1238, 1189, 1039, 998, 847, 748, 720, 689 cm⁻¹; ¹H NMR (400 MHz, CDCl₃, 40 °C): δ (ppm) 10.04 (br s, 1H), 7.13–7.25 (m, 7H), 7.04–7.12 (m, 2H), 6.66–6.71 (m, 2H), 6.13 (d, *J* = 8.0 Hz, 2H), 4.92 (s, 1H), 3.71–3.78 (m, 1H), 3.62 (t, *J* = 8.5 Hz, 1H), 3.26 (t, *J* = 8.5 Hz, 1H), 3.15–3.21 (m, 1H), 2.30–2.38 (m, 1H), 1.87–2.15 (m, 4H); ¹³C NMR (100 MHz, CDCl₃, 40 °C): δ (ppm) 151.4 (ddd, *J*_{C-F} = 253, 9.6, 3.8 Hz), 150.8 (ddd, *J*_{C-F} = 249, 9.6, 3.8 Hz), 145.4, 144.2 (q, *J*_{C-F} = 4.8 Hz), 143.9, 142.3 (q, *J*_{C-F} = 5.4 Hz), 139.2, (dt, *J*_{C-F} = 254, 15.3 Hz), 138.8, (dt, *J*_{C-F} = 252, 15.3 Hz), 137.3, 130.2, 129.5, 128.8, 128.7, 128.3, 117.0, 112.4–112.7 (m), 111.6, 111.2–111.5 (m), 80.5, 80.2, 60.8, 51.6, 50.9, 27.6, 22.9; Anal. Calcd for C₃₁H₂₄F₆N₂O: C, 67.14; H, 4.36; N, 5.05. Found: C, 67.10; H, 4.57; N, 4.88.

4.3.11. Bis(perfluorophenyl)(2-((3*R*,7*aS*)-2-phenylhexahydro-1*H*-pyrrolo[1,2-*c*]imidazol-3-yl)phenyl)methanol 1k

White solid; mp: 165–166 °C; $[\alpha]_D^{23} = +144.0$ (c 0.94, CHCl₃); IR (KBr): $\nu_{\max}$ 3441, 3054, 2936, 2846, 1648, 1599, 1524, 1485, 1355, 1330, 1302, 1190, 1156, 1128, 1095, 1007, 995, 979, 838, 820, 768, 753, 707, 697 cm⁻¹; ¹H NMR (500 MHz, CDCl₃): δ (ppm) 10.67 (br s, 1H), 7.17–7.32 (m, 5H), 6.96 (br d, *J* = 6.6 Hz, 1H), 6.73 (t, *J* = 7.3 Hz, 1H), 6.30 (d, *J* = 7.9 Hz, 2H), 5.34 (s, 1H), 3.84 (q, *J* = 8.3 Hz, 1H), 3.68 (t, *J* = 8.3 Hz, 1H), 3.34 (t, *J* = 8.3 Hz, 1H), 3.07–3.13 (m, 1H), 2.46 (q, *J* = 9.1 Hz, 1H), 1.88–2.13 (m, 4H); ¹³C NMR (126 MHz, CDCl₃): δ (ppm) 146.2–146.7 (m), 145.5, 144.2–144.7 (m), 141.7–142.2 (m), 140.6, 139.7–140.2 (m), 138.7–139.3 (m), 137.0, 136.7–137.2 (m), 129.4, 128.9, 128.8, 128.7, 127.8, 120.0–120.4 (m), 119.2–119.5 (m), 117.3, 111.4, 80.5, 79.8, 60.6, 51.8, 51.2, 27.6, 22.8; Anal. Calcd for C₃₁H₂₀F₁₀N₂O: C, 59.43; H, 3.22; N, 4.47. Found: C, 59.55; H, 3.35; N, 4.43.

4.4. Experimental procedure for X-ray crystallographic analysis of 1,4-amino alcohol 1a

A single crystal of **1a** was obtained by cooling a hot ethyl acetate solution of **1a** and was mounted on a glass fiber. All measurements were made on a Rigaku Mercury70 diffractometer using graphite monochromated Mo-K α radiation (λ = 0.71069 Å). The data were collected at a temperature of -50 ± 1 °C to a maximum

2θ value of 61.2° . A total of 744 oscillation images were collected. The crystal-to-detector distance was 45.00 mm. Readout was performed in the 0.137 mm pixel mode.

Of the 13598 reflections that were collected, 3943 were unique ($R_{\text{int}} = 0.0606$); equivalent reflections were merged. Data were collected and processed using CrystalClear(Rigaku).¹³ The linear absorption coefficient, μ , for Mo-K α radiation is 0.725 cm^{-1} . A numerical absorption correction was applied which resulted in transmission factors ranging from 0.984 to 0.991. The data were corrected for Lorentz and polarization effects.

The structure was solved by direct methods (SIR97)¹⁴ and expanded using Fourier techniques. The non-hydrogen atoms were refined anisotropically. Hydrogen atoms were refined using the riding model. All calculations were performed using the CrystalStructure crystallographic software package.^{15,16}

Crystal data for **1a**: $\text{C}_{31}\text{H}_{30}\text{N}_2\text{O}$, $M = 446.59$, monoclinic, $a = 9.866(4) \text{ \AA}$, $b = 8.373(3) \text{ \AA}$, $c = 15.449(6) \text{ \AA}$, $\beta = 105.144(5)^\circ$, $V = 1231.9(8) \text{ \AA}^3$, space group $P2_1$ (no. 4), $Z = 2$, $D_c = 1.204 \text{ g cm}^{-3}$, $F(000) = 476.00$, $T = 223(1) \text{ K}$, $\mu(\text{Mo-K}\alpha) = 0.725 \text{ cm}^{-1}$, 10735 reflections measured, 5694 independent ($R_{\text{int}} = 0.0481$). The final refinement converged to $R_1 = 0.0749$ for $I > 2.0\sigma(I)$, $wR_2 = 0.2009$ for all data.

CCDC 1449616 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

4.5. Typical experimental procedure for the enantioselective addition of diethylzinc to aldehydes

To a toluene (1.3 mL) solution of **1g** (58 mg, 0.10 mmol) under an atmosphere of argon was added a hexane solution of diethylzinc (1.1 M, 1.8 mL) through a syringe at 0°C , and the mixture was stirred at room temperature for 30 min. To the mixture was added a toluene (2.3 mL) solution of benzaldehyde (106 mg, 1.0 mmol) and the reaction mixture was stirred at room temperature for 5 h. Saturated ammonium chloride solution was then added to the reaction mixture. The organic layer was separated and the aqueous layer was extracted with dichloromethane three times. The combined organic layer was washed with water and brine, and dried over anhydrous Na_2SO_4 . After removal of solvent under reduced pressure, the crude product was purified by silica-gel column chromatography (Chromatorex NH, hexane/ $\text{CH}_2\text{Cl}_2 = 1:1$) to give (*S*)-1-phenyl-1-propanol (128 mg, 94%). The ee was determined to be 97% by HPLC analysis using a chiral column (Daicel Chiralcel OD-H ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 97/3; flow rate, 0.5 mL/min; t_r 28.3 min for minor peak, 33.4 min for major peak). The other reactions were carried out according to the typical procedure. The ees were determined by HPLC analyses using chiral columns. 1-(2-Bromophenyl)-1-propanol: Daicel Chiralcel OD-H ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 99/1; flow rate, 0.5 mL/min; t_r 40.8 min for minor peak, 43.0 min for major peak. 1-(1-Naphthyl)-1-propanol: Daicel Chiralcel OD-H ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 90/10; flow rate, 1.0 mL/min; t_r 10.3 min for minor peak, 8.7 min for major peak. 1-(2-Methoxyphenyl)-1-propanol: Daicel Chiralcel OB ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 90/10; flow rate, 0.5 mL/min; t_r 15.9 min for minor peak, 11.1 min for major peak. 1-(4-Methoxyphenyl)-1-propanol: Daicel Chiralcel OB ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 90/10; flow rate, 0.5 mL/min; t_r 24.9 min for minor peak, 18.7 min for major peak. 1-Cyclohexyl-1-propanol (as the corresponding 4-methoxybenzoate): Daicel Chiralcel AS-H ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 99.9/0.1; flow rate, 0.5 mL/min; t_r 18.2 min for minor peak, 20.6 min for major

peak. (*E*)-1-Phenyl-1-penten-3-ol: Daicel Chiralcel OD-H ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 97/3; flow rate, 1.0 mL/min; t_r 21.4 min for minor peak, 40.1 min for major peak. 1-Phenyl-3-pentanol: Daicel Chiralcel OD-H ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 99.5/0.5; flow rate, 0.3 mL/min; t_r 46.1 min for minor peak, 51.4 min for major peak. 1-Phenyl-1-ethanol: Daicel Chiralcel OD-H ($25 \text{ cm} \times 0.46 \text{ cm i.d.}$); eluent, hexane/*i*-PrOH = 95/5; flow rate, 0.5 mL/min; t_r 17.1 min for minor peak, 19.9 min for major peak.

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